Specification Approval Sheet

产品规格确认书

CELL Type: GRP1240GR-3.85V-55mAh

电芯型号: GRP1240GR-3.85V-55mAh

Prepared By/Date 编 制/日 期	Checked By/Date 审 核/日 期	Approved By/Date 批 准/日 期
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Customer Approval 客户批准	**Company Name(公司名称)**
	Signature/Date(签名/日期)
	Company Stamp(公司印章)

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文件修订记录

Revision History

版本 Version	修订内容 Contents	日期 Date	制定 Prepared	审核 Checked
A	新版发行 New release	2024-07-12	曾亮	胡彦庆

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1. 适用范围/ Scope

此规格书适用于深圳格瑞普电池有限公司的锂离子扣式可充电电池产品。本规格书在以下章节规定了锂离子扣式可充电池产品、产品特点和性能、相应的测试条件和方法。

The specification is suitable for the performance of li-ion rechargeable button cell produced by the SHENZHEN GREPOW BATTERY CO., LTD. This specification is made to describe the product, product characteristics and performance, relevant measurement conditions and methods applied to the rechargeable Lithium button cell as specified in following details.

2. 产品类型及型号/ Description and Model

2.1 产品类型：锂离子可充电扣式电池

Description: Li-ion rechargeable button cell

2.2 执行标准：IEC62133/UN38.3

Executive standard：IEC62133/UN38.3

2.3 产品型号/MODEL

GRP1240GR-3.85V-55mAh

3. 主要技术参数/ Specification

3.1 主要技术参数/Specification

序号/NO.	项目/Item	参数
1	标称容量/Typical Capacity	55 mAh 标称容量：0.2CmA,4.4~3.0V@23℃±2℃ Typical Capacity:0.2CmA,4.4~3.0V@23℃±2℃
2	最小容量/ Minimum Capacity	53 mAh
3	标称电压/Nominal Voltage	3.85V
4	内阻/ Impedance	≤ 380 mΩ
5	重量/Weight	1.4±0.15g（Ref）
6	出货电压/ Delivery Condition	3.7~3.9V
7	储存温度/Storage Temperature	40℃~60℃ 小于 1 个月 /Less than 1 month
		20℃~40℃ 小于 6 个月 /Less than 6 month
		-20℃~20℃ 小于 12 个月 /Less than 12 month

3.2 充放电参数及使用条件/ Charge and discharge parameters and conditions of use

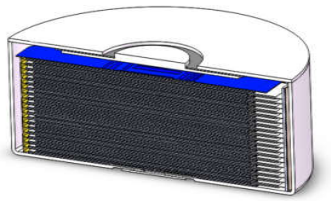
类别/ Category	电流/Current		电压 /Cut-off Voltage	温度 /Temperature
充电/Charge	标准充电/ Nominal Charge Current	0.5C	4.4V	0℃~45℃
	快速充电/Fast Charge	2C	4.2V	15℃~45℃
工作温度和电流/ Operating temperature and current	低倍率放电/ low Rate Discharge	0.2C	3.0V	-20℃-60℃
	标准放电/ Nominal Discharge Current	0.5C	3.0V	0℃~60℃
	快速放电/Fast Discharge	1C	3.0V	15℃~60℃

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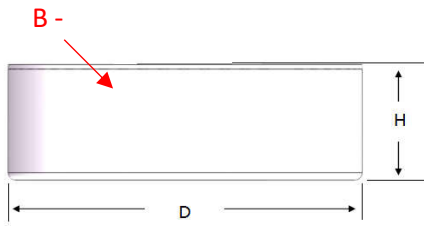
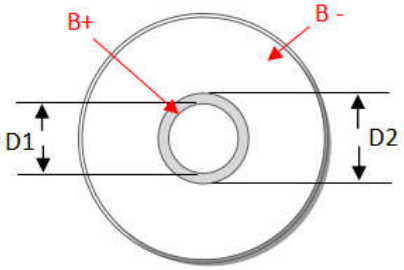
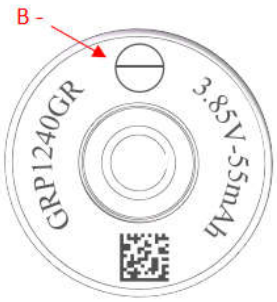
4. 电芯结构与尺寸示意图/ Structure and Dimension、QR information

4.1 电芯结构图/ Cell structure diagram

叠片工艺，壳盖全密封工艺/ Stacking and Shell cover fully sealed process.



4.2 外观示意图/ Appearance diagram

主视图/ Main view	俯视图/Top view	仰视图/ Bottom view
		
电芯高度 H（mm）： $4.1^{-0.2}$	**D1（mm）**: 3.0±0.1	
电芯直径 D（mm）： $12.1^{-0.1}$	**D2（mm）**: 4.0±0.1	

4.3 二维码信息/QR information

二维码包含 12 位字符/QR code contains 12 characters,
包含如下信息/Contains the following information:



H A 19 1 01 00001

Year	Month	Day	Scheduling code	Production line NO.	Serial No.
2022 H	Jan A	19			
2023 I	Feb B				
2024 J	Mar C				
… …	… …				

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5. 电芯性能测试/ Battery Cell Performance Test

5.1 测试环境及其它要求/ Test environment and other requirements

环境温度/Ambient temperature: 25℃±3℃

相对湿度/Relative Humidity: ≤75%RH

在进行下例各项测试前每颗电池应用 0.5C 放至 3.0V。如果没有特别规定，测试应在电池交付 1 个月内按以下各项条件进行:

Before proceed the following tests, the cells should be discharged at 0.5C to 3.0V cut-off. Unless otherwise stated, tests should be done within one month of delivery under the following conditions:

5.2 标准充放电制式如下/ Standard Charge/Discharge Conditions

充电: 以 0.5C 电流恒流充电至限制电压 4.4V 时，改为恒压充电，直到截止电流为 0.02C 停止充电，休息 5~10min，0.5C 恒流放电，截止电压 3.0V。

Charge: The battery will be charged to 4.4V with 0.5C from constant current to constant voltage, when the current is 0.02C, stop to charge, rest 5~10min. Discharge: 0.5C to 3.0V.

5.3 电性能测试/ Electrical performance test

测试项目/Test		单 位 Unit	标准/Criteria	条 件 Condition	备 注 Remarks
容量 Capacity		mAh	≥53	标准充放电 Standard Charge / Discharge	允许循环 3 次 Up to 3 cycles are allowed
开路电压 Open Circuit Voltage		V	≥4.35	标准充电后 1 个小时内 Within 1 hour after standard charge	颗 Unit cell
内阻 Internal Impedance (Ri)		mΩ	≤380	充满电后用 1kHz 交流内阻测试仪测试 Upon fully charge at 1kHz AC internal resistance tester test	*
不同温度下放电特性 Different Temperature Discharge Performance	常温 Normal Temperature		100%	标准充放电 Standard Charge / Discharge	3.0V/cell Cut-off
	高温 High Temperature		≥95%	标准充电后贮藏在60±2℃环境中2小时然后用 0.5C 放电 Standard charge Storage:2hrs at 60±2℃ 0.5C discharge at 60±2℃	
	低温 Low Temperature		≥60%	标准充电后贮藏在-20±2℃环境中 2 小时然后用 0.2C 放电 Standard charge Storage:4hrs at-20±2℃ 0.2C discharge at -20±2℃	

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储存性能 Storage Performance	常温 30 天 23℃/ 30days 23℃		剩余容量 Residual capacity ≥85% （初始容量 First Capacity）	标准充满电后 23 度贮藏 30 天，标准 0.5C 放电，所得容量为剩余容量 Standard full charge， storage at 23 degree for 30days， Standard discharge (0.5C)to test residual capacity	3.0V/cell Cut-off
			恢复容量 Recovery Capacity ≥95% （初始容量 First Capacity）	按标准充放电制式循环 3 次，取最大值为恢复容量 Standard charge/discharge for 3 cycles，to test recovery capacity	
	60℃4 小时/ 60℃ 4hrs		剩余容量 Residual capacity ≥95% （初始容量 First capacity） recovery	标准充电后贮藏在 60±2℃环境中 4 小时然后常温下 0.5C 放电，所得容量为剩余容量 Standard charge, Storage:4hrs at 60±2℃ 0.5C discharge at 25±3℃to test residual capacity	
			恢复容量 Recovery capacity ≥95% （初始容量 First capacity）	按标准充放电制式循环 3 次，取最大值为恢复容量 Standard charge/discharge for 3 cycles，to test recovery capacity	
寿命测试 Cycle Life Test		Cycle	≥500	充电：0.5C 充电至 4.4V，0.05C 截止，搁置 5min，放电：0.5C 放电至 3.0V，搁置 5min，当放电容量降至 0.5C 容量的 80%时，所完成的循环次数定义为该电芯的循环寿命 Charge:0.5C to4.4V 0.05C cut-off rest 5 min ,Discharge:0.5C to 3.0V, rest 5min, 80% or more of 1st cycle capacity at 0.5C discharge of operation	25℃±3℃

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6. 安全性能测试/ Safety Performance Test

序号/NO.	测试项目/Test		标准/Criteria	条件/Condition
1	外部短路 External short-circuit	常温/Room Temp. 25±5℃	不着火不爆炸 No fire and no explosion	标准测试环境下将电池充满电，然后接热电偶并（固定在电池侧面）置于通风的试验设备中，待电池温度达到测试温度后以阻值 80±20 mΩ的导体短接，当电池端电压＜0.2V 或电池表面温度达到环境温度±10℃时结束试验 Fully charge the cell in a standard test environment, then connect the thermocouple (fixed on the side of the cell) and place it in a ventilated test equipment. After the cell temperature reaches the test temperature, short-circuit it with a conductor with a resistance of 80±20 mΩ. The test ends when the voltage is less than 0.2V or the battery surface temperature reaches the ambient temperature ±10°C.
		高温 / High Temp. 55±5℃		
2	过充 Abnormal Charging Test		不着火不爆炸 No fire and no explosion	标准测试环境下将电池以 0.2C 放电至终止电压，然后接热电偶并（固定在电池侧面）置于通风的试验设备中，室温下以 3 C 充电至 4. 8V（或 4.8V 以上）截止，电池持续充电时间≥7h 或电池表面温度下降到比峰值低 20%时结束试验。 Under the standard test environment, discharge the cell at 0.2C to the final voltage, then connect the thermocouple and place it in a ventilated test equipment (fixed on the side of the battery), and charge it at 3C to 4.8V (or above 4.8V) at room temperature, the cell continues to charge for ≥7h or the cell surface temperature drops to 20% lower than the peak value, and the test ends.
3	强制放电 / Forced discharge		不着火不爆炸 No fire and no explosion	标准测试环境下将电池以 0.2C 放电至终止电压，然后接热电偶并（固定在电池侧面）置于通风的试验设备中，室温下然后以 1C 的电流对电池进行反向充电（恒流电源，设置 1C/5V），要求充电时间不低于 90min。 Under the standard test environment, the battery was discharged at 0.2C to the final voltage, then connected to a thermocouple and placed in a ventilated test equipment (fixed on the side of the battery), and then reversely charged with a current of 1C at room temperature (constant current power supply). , set 1C/5V), the charging time is not less than 90min.

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4	热冲击/ Thermal Shock Test	不着火不爆炸/ No fire and no explosion	标准测试环境下将电池充满电，将电池放于热箱中，然后接热电偶（固定在电池侧面），温度以（5±2℃）/min 的速率升至 130±2℃并保温 10min。 Fully charge the cell under the standard test environment, put the cell in the hot box, then connect the thermocouple (fixed on the side of the cell), the temperature rises to 130±2°C at a rate of (5±2°C)/min and keep it for 10min
5	重物冲击/Impact	不着火不爆炸/ No fire and no explosion	标准测试环境下将电池充满电，室温下电池置于一平面上，并将一个直径 15.8mm±0.1mm 的钢柱置于电池中心， 电池的纵轴平行于平面，让重量 9.1±0.46Kg 的重物从 610mm±25mm 高度自由落到电池中心上方的钢柱上，测试完毕观察 6h。 Fully charge the cell under the standard test environment, place the cell on a flat surface at room temperature, and place a steel column with a diameter of 15.8mm±0.1mm in the center of the cell. The heavy object falls freely on the steel column above the center of the battery from a height of 610mm±25mm, and the test is completed and observed for 6 hours.
6	平面挤压/ Crush	不着火不爆炸/ No fire and no explosion	标准测试环境下将电池充满电，室温下将电池水平放置于试验设备中，挤压的压力由一个直径为 32mm 活塞的液压油缸提供，挤压将一直持续到压力为 13 kN，一旦这个最大压力达到马上卸压。电池在最大面上承受一次挤压，然后搁置 4h 以上。 The cell is fully charged under the standard test environment, and the cell is placed horizontally in the test equipment at room temperature. The squeeze pressure is provided by a hydraulic cylinder with a diameter of 32mm. The squeeze will continue until the pressure reaches 13 kN. Once this maximum pressure Relieve pressure immediately. The battery is squeezed once on the largest side, and then left for more than 4 hours.

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7. 环境及机械性能测试/ Environmental and Mechanical Properties

序 号 /NO.	测试项目 Test	标准/Criteria	条 件/Condition
1	恒定湿热 / Steady damp-heat	无变形、无锈蚀、不泄气、不破裂、不起火、不爆炸，放电时间≥180min。/ NO deformation, NO rust, NO deflation, NO cracking, NO fire NO explosion, the discharge time shall be ≥ 180min.	标准测试环境下，按标准充电方法充满电，将电池放入 40±2℃，相对湿度为 90%~95%的恒温恒湿箱中搁置 48h，取出电池常温搁置 2h，目测电池外观，并以 0.2C 放电至 3.0V./ Under the standard test environment, fully charge the cell according to the standard charging method, put the cell in a constant temperature and humidity box with 40 ± 2 ℃ and relative humidity of 90%~95% for 48h, take out the cell and put it aside at room temperature for 2h, visually inspect the appearance of the battery, discharged by 0.2C to 3.0V.
2	温度循环 Temperature cycle	不起火、不爆炸、不泄气、不漏液 No fire, No explosion, No deflation, No leakage	标准充放电，放电容量为初始容量， 搁置 60 分钟，测量电压、内阻、厚度；然后进行下面步骤：， 1）温度 75±2℃，保温 6 h；2）降温至－40±2℃，保温 6 h；3）温度转换时间≤30min；4）重复步骤 1)~2），共循环 10 次。 结束后，搁置 2h，测量电芯/电池电压、内阻、厚度，以 0.5C 放电 3.0V，记为剩余容量；标准充放电 3 次，取放电容量最高一次作为恢复容量。 Standard charge and discharge, the discharge capacity is the initial capacity, put it on hold for 60 minutes, measure the voltage, internal resistance, and thickness; then perform the following steps:, 1) The temperature is 75±2°C, and the temperature is kept for 6 hours; 2) The temperature is lowered to -40±2°C, and the temperature is kept for 6 hours; 3) The temperature conversion time is less than or equal to 30min; After the end, put it on hold for 2h, measure the cell/battery voltage, internal resistance, thickness, discharge 3.0V at 0.5C, record it as the remaining capacity; standard charge and discharge 3 times, take the highest discharge capacity as the recovery capacity.

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3	高空模拟 / High altitude simulation	不起火、不爆炸、不泄气、不漏液 No fire, No explosion, No deflation, No leakage	标准测试环境下将电池充满电，记录电池电压、内阻、重量，将电池放入真空箱中。真空箱密闭后，逐渐减少其内部压力至不高于11.6kPa（模拟海拔 15240m）并保持 6 h，记录电池电压、内阻、重量 Fully charge the battery in a standard test environment, record the battery voltage, internal resistance, and weight, and place the battery in a vacuum box. After the vacuum box is sealed, gradually reduce its internal pressure to no higher than 11.6kPa (simulated altitude 15240m) and keep it for 6 hours, record the battery voltage, internal resistance, weight
4	振动/ Vibration	不起火、不爆炸、不泄气、不漏液 No fire, No explosion, No deflation, No leakage	将标准充电后的电芯固定在振动台上，沿 X、Y、Z 三个方向各振动 30 分钟，振幅 1.6 mm，振动频率为 10Hz~55Hz，每分钟变化为 1Hz。 After standard charge, fixed the cell to vibration table and subjected to vibration cycling that the frequency is to be varied at the rate of 1Hz per minute between 10Hz~ 55Hz, the excursion of the vibration is 1.6mm.The cell shall be vibrated for 30 minutes per axis of XYZ axes.
5	自由跌落 / Free falling(drop)	不着火不爆炸/ No fire and no explosion	将标准充电后的电芯从 1m 高度（最低点高度）位置自由跌落到置于水泥地面上的 18～20 ㎜厚的硬木板上，从 X、Y、Z 方向，每个方向自由跌落 1 次；结束后将电池以(0.5C)电流放电至 3.0V 测试放电容量。 Freely drop the standard charged battery cell from a height of 1m (the lowest point height) to a 18-20mm thick hardboard placed on the cement floor, and freely drop 1 time in each direction from X, Y, and Z directions. ; After the end, discharge the battery with (0.5C) current to 3.0V to test the discharge capacity.

8. 贮存和运输/ Storage and Transport

8.1 贮存要求/ Storage Requirement

电池按制定的要求进行贮存，带电量应为标准容量的 30%，开路电压 OCV 约为 3.8V，长期贮存时，必须保证电池电压不能低于放电截止电压。

The cells shall be stored within a proper temperature range as specifed in the Data Sheet. The state of charge shall be 30 % of the nominal capacity; open circuit voltage OCV about 3.8 V. When stored for a long time, care has to be taken that the battery voltage does not drop below the cut-off voltage.

项目/Items	要求/ Requirement	备注/Remark
湿度/Humidity	≤75%RH	/
环境温度/Ambient Temperature	-20~35℃	/
电池电压/Cell Voltage	3.6~3.8V	长期/for long time

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8.2 运输/transport

电池应在半荷电状态（30%~50%充电状态）下包装成箱进行运输， 在运输过程中应防止剧烈振动、冲击或挤压， 防止日晒雨淋， 适应于海陆空的交通运输。

Batteries should be packed into cartons under half-charged state (30%~50% charged state) for transportation. During transportation, severe vibration, impact or extrusion should be prevented, and the sun and rain should be protected. It is suitable for transportation by sea, land and air

9. 电池组装和焊接/ Battery Assembly and Soldering

9.1 电池连接/Cell Connection

严禁将导线或其他类型的连接器直接焊接到电池上。正确的电池连接只能由电池制造商自己完成。如果将导线或其他类型的连接器直接焊接到电池上的操作不是由电池制造商进行的，则有关保修、性能和安全的所有索赔都将被忽略。

应用外壳的超声波焊接:可以将塑料盖超声波焊接到塑料外壳上。但是，焊接时应避免将超声波功率直接施加到电池上。否则可能对电芯造成严重损害。

应用中的短路预防: 接线与电池之间应使用足够的绝缘层，以保持多重安全保护。电池外壳的设计应防止电池组装和设备使用过程中的短路。这是因为短路可能会导致产生烟雾或火灾。

Soldering or welding of wires or other types of connectors directly to the cell is strictly prohibited. A proper cell connection can only be done by the cell manufacturer itself. If soldering or welding of wires or other types of connectors directly to the cell will be done not by the cell manufacturer, all claims regarding warranty, performance and safety will be omitted.

Ultrasonic Welding of Application Housing: Ultrasonic welding of plastic lid to the plastic casing can be applied. However, the welding shall be done avoiding the application of ultrasonic wave power directly to the cells. Otherwise it may cause serious damage to the cells.

Prevention of Short-Circuit in Application： Enough insulation layer(s) between wiring and the cells shall be used to maintain multiple safety protection.The battery housing shall be designed to prevent shortcircuits while cell is assembled and during usage of device. This is because short circuits may cause generation of smoke or fire.

9.2 电池组装/Cell Assembly

重要提示：始终避免电池外壳与尖锐物体、角落或点的任何可能接触可能会刺穿或破坏电池，避免在组装中对电池本身施加机械应力（张力，压力）。请勿从格瑞普提供的电池中移除或拆卸任何组件。不要让电池承受比提供的数据表中指定的温度更高的温度。不要让电芯受到超声波焊接过程的振动或能量。避免在组装和修整过程中电池意外短路。

避免在组装和修整过程中对电池造成意外机械损坏。电池的包装必须由绝缘材料制成，避免放电或短路。

Important: Always avoid any possible contact of cell housing with sharp objects, corners, or points which could puncture or damage the cell. Avoid applying mechanical stress (such as tension, pressure) to cell itself during assembly. Do not remove or disassemble any component from the original GREPOW supply confguration. Do not subject cell to higher temperatures than specifed in Data Sheet provided. Do not subject cell to ultrasonic weld process vibration or energy. Avoid accidental short-circuit of cell during assembly and fnishing processes.

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Avoid accidental mechanical damage to cell during assembly and fnishing processes. Packaging for cell has to be made of insulating material, avoiding discharge or short-circuiting.

9.3 焊接要求/ Soldered Requirements

使用组装好的带有引出片或导线的电池，可以焊接到 PCB 上。为此，请考虑以下有关纽扣电池可焊性的指南

With assembled tags or wires, the button cell may be soldered onto a PCB. For this purpose, Please consider the following guidance on the solderability of the button cell.

9.3.1 烙铁使用/ Using a Soldering Iron

不要让烙铁直接接触电池主体。在 3 秒内快速进行焊接，同时将烙铁头温度保持在 320 ºC 左右，不要让池体温度超过 60 ºC

Do not allow the soldering iron to make direct contact with the body of the cell. Proceed with soldering quickly within 3 seconds while maintaining the iron tip temperature at about 320 ºC, and do not allow the temperature of the cell body to exceed 60 ºC.

9.3.2 浸焊 / 波峰焊/Dip Soldering / Wave Soldering

这两种焊接程序可能会使电池短路。这将对电芯造成不可逆转的损害。这将极大地损害性能，并可能导致安全风险。

These two soldering procedures could short circuit the battery. This will cause irreversible damage to the cell. This will impair performance dramatically and might lead to safety risks as well.

9.3.3 回流焊接/ Reflow Soldering

切勿使用回流焊接。在标准的回流过程中，身体和电池内部的温度会上升到一定程度，这将对电池造成不可逆转的损害。这将极大地损害性能，并可能导致安全风险。由于温度过高，在回流焊接过程中存在爆炸 / 爆裂和电解液泄漏的风险。

NEVER USE REFLOW SOLDERING，During a standard reflow process the temperature on the body and inside the cell will rise to a level which will cause irreversible damage to the cell. This will impair performance dramatically and might lead to safety risks as well. There is a risk of explosion / bursting and electrolyte leakage during reflow soldering due to excessive temperatures.

10. 处理注意事项和禁止事项/ Handling Precautions and Guideline

本章节介绍了操作注意事项和禁止事项。如果您对任何一点有任何疑问，请联系我们。

In this section there is information about the handling precautions and prohibitions for the button cell series. If you have any questions regarding any point please consult us.

10.1 主要信息/General Information

锂电池提供高能量密度，通常与高倍率能力相结合，以造福客户。由于这些优异的性能特性，锂电池存在一定的安全风险。如果短路，可能会产生热量，有时可能会产生火花。超出推荐限值的不当处理可能导致气体产生、泄漏和火灾。

Lithium batteries provide a high energy density which is often combined with a high rate capability to the beneft of the customer. Due to these excellent performance properties, Lithium batteries contain a certain safety risk. If short-circuited, heat and sometimes sparks may be generated. Mistreatment beyond the recommended limits can cause gas generation, leakage and fire.

如果客户需要本文档中描述的应用或操作条件以外的其他应用或操作条件，请提前与我们联系。在这种情况下，可能需要额外的测试和实验来验证在这种条件下的性能和安全性。GREPOW 不对安全、性能、功能、兼容性或特定用途的适用性负责，除非这些功能已在规范中明确传达和描述。

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The customer is requested to contact us in advance, if and when the customer needs other applications or operating conditions than those described in this document. In this case additional tests and experiments may be necessary to verify performance and safety under such conditions. GREPOW buutton cell shall not be responsible for safety, performance, functionality, compatibility or fitness for a particular purpose unless such features have been expressly communicated and described in the specifcation.

如果电池在本指南所述条件之外的其他条件下使用，GREPOW 纽扣电池对任何事故概不负责。如果认为有必要，GREPOW 纽扣电池将书面通知客户有关正确使用和处理电池的改进，未经 GREPOW 事先书面批准，请勿修改和/或打开电池或电池。

GREPOW button cell will take no responsibility for any accident when the cell is used under other conditions than those described in this guideline. GREPOW button cell will inform, the customer in writing of improvement(s) regarding proper use and handling of the cell, if deemed necessary, Do not modify and or open cells or batteries without prior written approval by GREPOW

10.2 充电/Charge

10.2.1 充电电流/Charge current

充电电流不应超过数据表中规定的最大充电电流。以高于推荐的电流充电可能会损坏电池性能和安全特性，并可能导致发热或泄漏.

Charging current should not exceed maximum charge current specifed in the Data Sheet.
Charging with higher current than recommended may cause damage to cell performance and safety features, and can lead to heat generation or leakage

10.2.2 充电电压/Charge voltage

严禁在绝对最大电压 4.4 以上充电。充电器应按此条件设计。仅使用指定的充电器。以高于规定的电压充电可能会损坏电池性能和安全特性，并可能导致发热、发热或电池泄漏。

Charging at above 4.4, which is the absolute maximum voltage, is strictly prohibited. The charging has to be done according to the data sheet. The charger shall be designed in accordance with this condition. Use specifed charger only. Charging with higher voltage than specifed may cause damage to cell performance and safety features, and can lead to fre, heat generation or cell leakage.

10.2.3 充电温度/Charge temperature

电池应在数据表中规定的温度范围内充电。如果电池在超出指定范围的温度下充电，可能会发生泄漏、发热或其他损坏。即使在规定的温度范围内，高低温反复充放电也可能导致电池性能下降。

The cell shall be charged within the range of specifed temperatures in the Data Sheet. If the cell is charged at a temperature out of the specifed range, leakage, heat generation, or other damages may occur. Repeated charging and discharging at high and low temperature may cause degradation of cell performance even within the specifed temperature range.

10.2.4 截止反向充电/ Prohibition of Reverse Charging

禁止反向充电。电池应正确连接。在连接任何电线之前，必须确认极性。
反向充电会对电池造成损坏，并导致电池性能和电池安全性下降（包括发热或泄漏）

Reverse charging is prohibited. The cell shall be connected correctly. The polarity has to be confrmed before connecting any wires. Reverse charging will cause damage to the cell(s) and will lead to a loss of cell performance and cell safety (including heat generation or leakage).

10.2.5 禁止涓流充电或连续充电/ Prohibition of Trickle Charging or Continuous Charging

禁止涓流充电或连续充电。涓流充电或连续充电会导致过度充电、产生内压和电池退化。电池应以恒定电流充电至 4.40 V，然后以恒定电压和逐渐减小的电流充电。在大约 0.02C 电流必须停止充电。仅当电池已释放出可观的容量，或电池电压已降至 4.0 V 以下时，才应重新开始充电。

Trickle charging or continuous charging is prohibited.Trickle charging conditions or continuous charging can lead to overcharging, generation of internal pressure and degeneration of the cell. The cell shall be charged with constant current until 4.40 V, then with constant voltage and tapering current. At approx. 0.02C current the charging must be stopped. Charging should restart only if appreciable capacity has been discharged from the cell, or the cell voltage has fallen itself below a voltage level of 4.0 V

10.3 放电/Discharge

10.3.1 放电电流/Discharge current

电池的放电电流应小于或等于数据表中规定的最大放电电流。高放电电流可能会显着降低放电容量，或导致过热。

The cell shall be discharged at less or equal than the maximum discharge current specifed in the Data Sheet. High discharge current may reduce the discharging capacity signifcantly, or cause overheating

10.3.2 放电温度/Discharge temperature

电池应在制定的温度范围内放电。

The cell shall be discharged within the temperature range that is specifed..

10.3.3 过放电/ Over-Discharging

长时间不使用电池可能会导致过放电。为防止过度放电，电池应定期充电以保持电压在 3 V 至 3.8 V 范围内。过度放电可能会导致电池性能下降或损坏电池功能。实际应用中应配备一个装置防止指定的截止电压以下进一步放电。

PCM 过放检测阈值 / 电压不得作为电池的截止电压。

过放电时，电池应以低电流（0.02 – 0.05C）启动 15~30 分钟，即预充电，在快速充电开始之前。根据数据表充电应在 15 - 30 分钟内单节电池电压升至约 3 V 以上后开始，可通过使用适当的预充电定时器来确定和控制。如果单节电芯电压在预充电时间内没有上升到 3V 左右，充电器应具有停止进一步连续充电并显示电芯处于异常状态的功能。

Not using the cell for a long time may lead to overdischarge. In order to prevent overdischarging, the cell shall be charged periodically to maintain a voltage in the range of 3 V to 3.8 V. Over-discharging may cause loss of cell performance, or damage battery function. The application device shall be equipped with a device to prevent further discharging below the cutoff voltage specifed.

PCM over-discharge detection threshold / voltage must not be used as cutoff voltage for battery.

Also the charger shall be equipped with a device to control the recharging procedures as follows:

In case of over-discharging, the cell / battery pack shall start with a low current (0.02 – 0.05C) for 15 ~30 minutes, i.e. precharging, before rapid charging starts.The charging according to the Data Sheet shall be started after the individual cell voltage has risen above about 3 V within 15 – 30 minutes, which can be determined and controlled by the use of an appropriate timer for precharging. In case the individual cell voltage does not rise to about 3 V within the pre-charging time, the charger shall have functions to stop the further continuous charging and display that the cell(s) is / are in an abnormal state.

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10.3.4 保护电路模板（PCM）/ Protection Circuit Module (PCM)

电池应配备能正确保护电池的 PCM，例如：在充电控制电路出现故障的情况下。PCM 应具有 (1) 过充电防止、(2) 过放电防止和 (3) 过电流防止功能，以保持安全并防止电池性能显着下降。过电流可能因外部短路而发生。

The cell(s) shall be provided with a PCM which can protect cell properly, e.g. in case of failing Charge Control Circuit. PCM shall have functions of (1) overcharging prevention, (2) over-discharging prevention, and (3) over current prevention, to maintain safety and prevent signifcant deterioration of cell performance. The overcurrent can occur by external short circuit.

10.3.5 禁止过放电/Over-Discharge Prohibition

过放电防止功能单元应尽量减少耗散电流，以避免电池进一步下降电压低于 2.0 V。建议 PCM 的耗散电流应设计为在过放电防止功能激活后最小化为 0.5 微安或更小，以最大限度地减少对电池保质期的影响。如果单个电池电压低于 2.0 V PCM 应具有将电池与电子电路断开的功能，在任何情况下均不得对电池充电。

Over-discharge prevention function shall work to minimize dissipation current to avoid further drop in cell voltage below 2.0 V. It is recommended that the dissipation current of PCM shall be designed to be minimized to 0.5 microamperes or less after the over-discharge prevention function activates in order to minimize effects on shelf life of the battery. In case the individual cell voltage falls below 2.0 V PCM shall have functions to disconnect the cell from electronic circuit and cell shall not be recharged in any case.

10.4 应用/ Application

最终产品随附的说明手册应包含以下警告声明：

(1) 必须由经过培训的技术人员更换电池。只能使用带有（电池制造商名称或最终产品制造商名称）的电池进行更换。使用其它电池可能会导致火灾或爆炸。（最终产品名称）中使用的电池只能在（最终产品制造商）服务中心更换。”

(2) 注意：设备中使用的电池如果处理不当可能会引起火灾或化学灼伤。不要拆卸、加热到 100 ºC 以上或焚烧。”

(3) 根据当地法律法规正确处理废旧电池。远离儿童——如果是有害的吞了！”

(4) 警告：火灾、爆炸和烧伤风险。请勿拆卸、压碎、加热至 100 ºC 以上，短路或焚烧。

The instruction manual supplied with the end product shall contain the following warning notice:

(1) Replacement of battery has to be done by trained technician. For replacement only batteries with (Battery Manufacturer‘s name or endproduct manufacturer‘s name)may be used. Use of another battery may present a risk of fire or explosion.“ or „The battery used in the (End Product Name) must be replaced at (End product manufacturers) service center only.“

(2) Caution: The battery used in this device may present a fire or chemical burn hazard if mistreated. Do not disassemble, heat above 100 ºC or incinerate."

(3) Dispose of used battery properly considering local laws and rules. Keep away from children – harmful if swallowed!”

(4) WARNING: Risk of Fire, Explosion, and Burns. Do Not Disassemble, Crush, Heat above 100 ºC, Short-Circuit or Incinerate.

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10.5 禁止拆卸/ Prohibition of Disassembly

切勿拆卸电池，拆卸电池可能会导致电池内部短路，这可能会进一步导致放气、起火或其他问题。

Never disassemble the cells. Disassembling cells may cause an internal short-circuit in the cell, which could further cause gassing, fire, or other problems

10.6 有害的电解液/ Harmful Electrolytes

从电池中泄漏出来的电解质对人体有害。如果电解液接触到皮肤、眼睛或身体其他部位，应立即用水冲洗电解液。向医生寻求医疗建议。

An electrolyte which leaks out from the cells is harmful to the human body. If the electrolyte comes into contact with the skin, eyes or other parts of body, the electrolyte shall be flushed immediately with water. Seek medical advice from a physician.

10.7 禁止短路/ Prohibition of Short-Circuit

切勿使电池短路。它会产生非常高的电流，从而导致电池发热，这可能会导致电解液泄漏、放气或起火。

Never short-circuit the cells. It causes generation of very high currents resulting in heating of the cells, which may cause electrolyte leakage, gassing or fire.

10.8 禁止电池投入火中/Prohibition of Dumping of Cells into Fire

禁止电池投入火中。

Never incinerate nor dispose of cells into fire

10.8 禁止电池浸入液体中/ Prohibition of Cells Immersion into Liquid

如水，电芯不能被水、海水等液体、汽水、果汁、咖啡等饮料浸泡

Such as Water,The cells shall never be soaked with liquids such aswater, sea water, drinks such as soft drinks, juices, coffee or others.

10.10 禁止使用损坏的电池/Prohibition of Use of Damaged Cells

电池在运输过程中可能会因震动或其他原因而损坏。如果发现电芯出现异常，如：电芯不锈钢外壳损坏、变形、电解液有异味、电解液泄漏或其他异常，则不得继续使用。有电解液气味或泄漏的电池应远离火源，以免着火。

Cells may be damaged during shipping by shocks, or other causes. If any abnormal features of the cells are found such as: damage to the stainless steel housing of the cell, deformation of the cell container, smell of electrolyte, an electrolyte leakage, or other abnormalities, the cells shall not be used any more. Cells with a smell of electrolyte or leakage shall be kept away from fire to avoid ignition

10.11 电池仓设计/Battery Compartment Design

保护电路应与电池隔离，以减少因事故可能发生的任何电解液泄漏造成的损坏。电池舱的设计应不允许泄漏的电解液进入保护电路。选择电池外壳材料时，应考虑电池外壳材料对电解液的电阻。

在恶劣的条件下，电池可能会排气。必须小心，在任何滥用条件下都可以释放过压。组装不得干扰通风机制。在恶劣的条件下，电池可能会排气；确保安全通风，需要在轴向上最多留出 1.5 mm 的额外空间。

Protection circuit shall be isolated from the cell to diminish damage from any electrolyte leakage which may occur by mishap. The battery compartment shall be designed not to allow leaked electrolyte access to protection circuit. Battery case material resistance for electrolyte shall be considered when battery case material is selected.

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Under abusive conditions the cell may vent. Care has to be taken, that overpressure can be released in any abusive condition. Assembly must not interfere with venting mechanism. Under abusive conditions the cell may vent; to ensure safe venting, up to 1.5 mm of additional space in axial direction is necessary.

10.12 警告/Waring

以下警告语言将与电池所打包的信息一起提供，并且电池或使用它们的设备:

• 将被认为可吞食的小电池和电池放在儿童接触不到的地方。
• 吞咽可能导致烧伤、软组织穿孔和死亡。严重烧伤可在 2 小时内发生的摄取。
• 如果误食电池或电池，请立即寻求医疗帮助

The following warning language is to be provided with the information packaged with the cells and batteries or equipment using them:

• Keep small cells and batteries which are considered swallowable out of the reach of children.
• Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2 h of ingestion.
• In case of ingestion of a cell or battery, seek medical assistance promptly.

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